

ARSALAN ANWAR

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Work Experience

Data Scientist

Aug 2025 - Present

Medidata Solutions

New York, USA

- **Fine-tuned an LLM** for Edit Check Generation, implementing a **new SFT training and evaluation framework** on the Phi-4-mini base model, improving generation accuracy by **22%**
- Re-architected the User Requirements Generation and Deduplication pipeline by refactoring the embedding service into a **Ray-based microservice**, reducing end-to-end latency from **45 seconds** to under **2 seconds** (>95% improvement)
- Built automated stress testing and performance analysis **pipelines to evaluate system behavior** under normal and peak loads, delivering actionable metrics that increased system reliability and strengthened stakeholder confidence ahead of production rollout

AI Software Engineer

Jun 2025 - Nov 2025

Novum AI

New York, USA

- **Designed and deployed a real-time LLM-powered call center assistant** that generated context-aware sales recommendations from live transcriptions in **<2s latency**, improving agent decision speed and increasing conversion rates
- **Built and optimized a hybrid retrieval pipeline** using layout-aware hierarchical chunking, sparse BM25 search, dense vector search, and cross-encoder re-ranking, significantly improving contextual precision during live conversations
- **Implemented a Human-in-the-Loop learning framework** that logged model suggestions and agent-selected responses to update contextual memory & iteratively refined future recommendations based on user behavioral signals

Data Science Intern

Feb 2025 - Jun 2025

Medidata Solutions

New York, USA

- Developed an **automated validation framework** to evaluate model performance for **clinical form extraction** and **edit check generation**, significantly reducing manual QA efforts
- Co-designed a POC for the next-gen **ML Prediction Service (Coder+)** for **intelligent medical coding** using **RAG** architecture with **vector-embedded MedDRA** and **WHODrug** data stored in **Mongo Vector DB**

Data Science Intern

May 2024 - Aug 2024

Medidata Solutions

New York, USA

- Automated clinical trial consent workflows using **NLP** and **Large Language Models (LLMs)**, cutting setup time by **45%** and costs by **\$6 million** annually
- Achieved **98% accuracy** in form-field mapping using **AWS Bedrock** and **Claude AI**, reducing annotation time by **87%** (15min to 2min)

Data Scientist

Feb 2023 - Aug 2023

Course5 Intelligence

Bangalore, IN

- Boosted average revenue per customer by **15%** through market basket analysis (**Apriori**), enabling targeted upselling strategies for **50K+** users
- Reduced customer attrition risk by **20%** using a CLV-Churn ensemble ML model, effectively identifying high-value churn risks in the top **5%** segment, enhancing customer retention strategies

Data Scientist

Jan 2020 - Feb 2023

West Pharmaceutical Services

Bangalore, IN

- Designed a **XceptionNet** deep learning model to classify 15+ defect categories in rubber stoppers, achieving **92% accuracy** (**37%** improvement over manual processes). Containerized with **Docker** and deployed on **Azure**, reducing defect resolution time by **62%**
- Architected a document parsing system using **BERT** on **5 TB** of data, improving keyword-to-blog mapping by **23%**
- Built robust **ELT pipelines** on **Azure Data Factory** for 6+ data sources, reducing data preparation time by **40%**

Technical Skills

Programming: Python, SQL, C#, Java, C/C++, PySpark, Spark

Machine Learning Frameworks: Unsloth, TensorFlow, PyTorch, Keras, Scikit-learn, Numpy, OpenCV

NLP/LLMs: BERT, RAG, LangChain, Transformers

Cloud/MLOps: AWS, Azure, Docker, MLFlow, Azure Data Factory, Firebase, Pinecone, MongoDB

Data Visualization & Other Tools: Power BI, Tableau, Flask, Django, Postman, Git, Snowflake, Excel

Education

New York University

New York, USA

Master of Science in Computer Science

Sep 2023 - May 2025

Visvesvaraya Technological University

Bangalore, India

Bachelor of Engineering in Computer Science

Aug 2016 - Aug 2020